Lab 4: Complements to Prediction

ECON 470 — Microeconomics of Artificial Intelligence

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Groups: 2–3 students. **Only one submission per group, please.**

Submit: A writeup with your answers to the numbered questions below. Show your work for all calculations.

Points: 100 total. See point values next to each problem.

Time estimate: 2–3 hours. The handout does most of the model-building; the lab pushes you to apply it.

What you need: The handout *Judgment and Prediction: From Experimentation to Delayed and Uncertain Experimentation* (in hand from class), Gans Chapter 5 §5.1, and a calculator. No coding required.

If you get stuck: The Prediction Tutor can help you work through problems without giving away the answer: https://chatgpt.com/g/g-69d815bc1cdc81919933d59a1f8dd044-value-of-ai-prediction-tutor. Use it sparingly — you'll learn more by wrestling with the questions yourself first.

Pacing. Problems 1 and 2 use only Part I of the handout (instant experimentation), which we have already covered. Problem 3 uses Part II (delayed and uncertain experimentation), which we will cover on Monday. **You can start Problems 1 and 2 immediately — don't wait until after Lab 3 is in.**

Notation throughout this lab. All variables and assumptions are as defined in the handout: states $\theta \in \{1, 2\}$ each equally likely; safe action paying S; risky action paying R or r with $R > S > r$; common probability v that the risky action pays R in any given state; experimental quality λ; discount factor δ; assumptions A1 (Safe Default) and A2 (Judgment Insufficient). Running parameters are the handout's: $R = 100$, $r = 20$, $S = 75$, $v = 0.3$, $\delta = 0.95$.

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1 Setting Up the Model [25 pts]

(a) Verify Assumption A1 (Safe Default) by deriving the largest value of v for which $vR+(1-v)r \leq S$ with $R = 100$ and $r = 20$, $S = 75$. Confirm that the running parameter $v = 0.3$ satisfies it.

(b) Verify Assumption A2 (Judgment Insufficient) at the running parameters. State in one sentence what A2 rules out.

(c) Suppose A1 *failed* — that is, the risky action's prior expected payoff exceeded S. What would the agent do absent any prediction or experimentation? Why does Gans need A1 in place to make the complementarity story interesting? *(One sentence.)*

(d) Now consider the value $v = 0.5$. Does A1 still hold? Does A2 still hold? *(No need to recompute the model yet — just check the assumptions.)*

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2 The Actionability Mechanism [30 pts]

This problem builds on the Summary table at the end of Part I of the handout (the four cells $V_\emptyset, V_P, V_J, V_{PJ}$).

(a) From the handout, the marginal value of prediction is $V_P - V_\emptyset = 0$ when the agent has no experimentation, but $V_{PJ} - V_J = 5.25$ when the agent has experimentation. Explain in 2–3 sentences why one is zero and the other is positive — in terms of *which experimental outcomes change the action.*

(b) At the running parameters, the three experimental outcomes have probabilities $v^2 = 0.09$ (both states pay R), $2v(1-v) = 0.42$ (mixed), and $(1-v)^2 = 0.49$ (both states pay r). The mixed outcome is the *most likely* of the three, yet it contributes nothing to $V_J - V_\emptyset$. Explain in 1–2 sentences why.

(c) Imagine a "perfect" agent who has both prediction *and* free, perfect experimentation, and never makes a mistake. Compute their expected payoff. Then compute $V_{\text{perfect}} - V_\emptyset$. What share of this total surplus is captured by free experimentation alone (no prediction)? By prediction alone (no experimentation)? By the complementarity (the joint use of both, beyond the sum of the individual contributions)?

(d) In one sentence: what does the handout mean when it says prediction makes experimental results *actionable*?

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3 The Threshold and Comparative Statics [45 pts]

In Part II of the handout, experimentation is delayed (one-period discount δ) and may fail (success probability λ). The threshold for the agent to find experimentation worthwhile, *given prediction*, is

$$\bar{\lambda} = \frac{S(1-\delta)}{\delta v(R-S)}.$$

(a) Verify by direct computation that $\bar{\lambda} \approx 0.53$ at the running parameters. Show the substitution.

(b) Suppose v rises to 0.5 (the risky action is more likely to pay R). Recompute $\bar{\lambda}$. Does the threshold rise or fall? Give a one-sentence economic interpretation.

(c) Suppose instead δ rises to 0.99 (the agent is much more patient). Recompute $\bar{\lambda}$. Why does patience lower the threshold?

(d) Suppose instead R rises from 100 to 130 while S stays at 75 (bigger upside, $R - S$ wider). Recompute $\bar{\lambda}$. In one sentence: who has a stronger incentive to invest in prediction — a firm with a large $R - S$ or a firm with a small one?

(e) Take the partial derivatives $\partial\bar{\lambda}/\partial v$, $\partial\bar{\lambda}/\partial\delta$, and $\partial\bar{\lambda}/\partial(R-S)$. Sign each one and confirm that the signs match the directions you found numerically in (b)–(d).

(f) The handout shows that *without* prediction, the analogous threshold is

$$\bar{\lambda}_{\text{no pred}} = \frac{S(1-\delta)}{\delta v^2 (R-S)} = 1.75$$

at the running parameters — above 1, so the agent never experiments. Holding R, r, S, and δ fixed at the running values, find the largest value of v for which the agent *still* would not experiment without prediction (i.e., for which $\bar{\lambda}_{\text{no pred}} \geq 1$). Interpret your answer in one sentence: what does this tell you about the range of priors over which prediction is *necessary* (not just helpful) for experimentation to be worthwhile?